20260928 Quiz

for (MATH210) Applied Complex Variables

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Exercise 26.4. In each case, determine the singular points of the function and state why the function is analytic everywhere else:

(a) $f(z) = \dfrac{2z+1}{z(z^2+1)}$.

(b) $f(z) = \dfrac{z^3+i}{z^2-3z+2}$.

(c) $f(z) = \dfrac{z^2+1}{(z+2)(z^2+2z+2)}$.

Solution. Each function is rational, so by the differentiation rules it is analytic wherever its denominator is nonzero.

(a)
$$z(z^2+1) = z(z-i)(z+i),$$
so the singular points are
$$\boxed{z = 0,\ \pm i}.$$

(b)
$$z^2 - 3z + 2 = (z-1)(z-2),$$
so the singular points are
$$\boxed{z = 1,\ 2}.$$

(c) Since
$$z^2 + 2z + 2 = (z+1)^2 + 1 = (z+1-i)(z+1+i),$$
the singular points are
$$\boxed{z = -2,\ -1 \pm i}.$$

In each case the function is analytic at every other point.

□

Exercise 26.6. Use results in Sec. 24 to verify that the function

$$g(z) = \ln r + i\theta \qquad (r > 0,\ 0 < \theta < 2\pi)$$

is analytic in the indicated domain of definition, with derivative $g'(z) = 1/z$. Then show that the composite function

$$G(z) = g(z^2 + 1)$$

is analytic in the quadrant $x > 0$, $y > 0$, with derivative

$$G'(z) = \frac{2z}{z^2 + 1}.$$

Suggestion. Observe that $\mathrm{Im}(z^2 + 1) > 0$ when $x > 0$, $y > 0$.

Solution. For

$$g(z) = u(r, \theta) + iv(r, \theta), \qquad u = \ln r, \quad v = \theta,$$

we have

$$ru_r = 1 = v_\theta, \qquad u_\theta = 0 = -rv_r.$$

The relevant partial derivatives are continuous for $r > 0$, $0 < \theta < 2\pi$. Hence the theorem in Sec. 24 gives

$$g'(z) = e^{-i\theta}(u_r + iv_r) = \frac{e^{-i\theta}}{r} = \boxed{\frac{1}{z}}.$$

Now let $z = x + iy$ with $x > 0$ and $y > 0$. Since

$$z^2 + 1 = (x^2 - y^2 + 1) + 2xyi,$$

we have

$$\mathrm{Im}(z^2 + 1) = 2xy > 0.$$

Thus $z^2 + 1$ lies in the domain of the chosen branch of g. The inner function $z^2 + 1$ is entire, so the composition is analytic in the quadrant. By the chain rule,

$$\boxed{G'(z) = g'(z^2 + 1) \cdot 2z = \frac{2z}{z^2 + 1}}.$$

□

Exercise 26.7. Let a function f be analytic everywhere in a domain D. Prove that if $f(z)$ is real-valued for all z in D, then $f(z)$ must be constant throughout D.

Solution. Write

$$f(z) = u(x, y) + iv(x, y).$$

Since f is real-valued throughout D,

$$v(x, y) = 0$$

there, so $v_x = v_y = 0$. Because f is analytic, the Cauchy–Riemann equations give

$$u_x = v_y = 0, \qquad u_y = -v_x = 0.$$

Consequently

$$f'(z) = u_x + iv_x = 0$$

throughout D. By the theorem in Sec. 25, an analytic function whose derivative vanishes throughout a domain is constant. Hence f is constant on D.

□

Exercise 27.2. Let the function $f(z) = u(x, y) + iv(x, y)$ be analytic in a domain D, and consider the families of level curves $u(x, y) = c_1$ and $v(x, y) = c_2$, where c_1 and c_2 are arbitrary real constants. Prove that these families are orthogonal. More precisely, show that if $z_0 = (x_0, y_0)$ is a point in D which is common to two particular curves $u(x, y) = c_1$ and $v(x, y) = c_2$ and if $f'(z_0) \neq 0$, then the lines tangent to those curves at (x_0, y_0) are perpendicular.
Suggestion. Note how it follows from the pair of equations $u(x, y) = c_1$ and $v(x, y) = c_2$ that

$$u_x + u_y \frac{\mathrm{d}y}{\mathrm{d}x} = 0 \qquad \text{and} \qquad v_x + v_y \frac{\mathrm{d}y}{\mathrm{d}x} = 0.$$

Solution. At z_0, the Cauchy–Riemann equations give

$$u_x = v_y, \qquad u_y = -v_x.$$

Also,

$$|f'(z_0)|^2 = u_x^2 + u_y^2,$$

so $f'(z_0) \neq 0$ implies that u_x and u_y are not both zero.

Suppose first that $u_x u_y \neq 0$. Implicit differentiation gives the slopes

$$m_u = -\frac{u_x}{u_y}, \qquad m_v = -\frac{v_x}{v_y}.$$

Using the Cauchy–Riemann equations,

$$m_v = -\frac{-u_y}{u_x} = \frac{u_y}{u_x},$$

and therefore

$$m_u m_v = -1.$$

Thus the tangent lines are perpendicular.

If $u_y = 0$, then $u_x \neq 0$. The level curve of u has a vertical tangent, while the Cauchy–Riemann equations give $v_x = 0$ and $v_y = u_x \neq 0$, so the level curve of v has a horizontal tangent. The case $u_x = 0$ is analogous. Hence the two families are orthogonal wherever $f'(z_0) \neq 0$.

□

Exercise 29.1. Use the theorem in Sec. 28 to show that if $f(z)$ is analytic and not constant throughout a domain D, then it cannot be constant throughout any neighborhood lying in D.
Suggestion. Suppose that $f(z)$ does have a constant value w_0 throughout some neighborhood in D.

Solution. Suppose, to the contrary, that $f(z) = w_0$ throughout some neighborhood contained in D. Let

$$g(z) = w_0$$

for all $z \in D$. Then g is analytic in D, and f and g agree throughout that neighborhood. By the uniqueness theorem in Sec. 28,

$$f(z) = g(z) = w_0$$

throughout D. This contradicts the assumption that f is not constant on D. Therefore f cannot be constant throughout any neighborhood lying in D.

□

Exercise 29.5. Show that if the condition that $f(x)$ is real in the reflection principle (Sec. 29) is replaced by the condition that $f(x)$ is pure imaginary, then equation (1) in the statement of the principle is changed to

$$\overline{f(\overline{z})} = -f(z).$$

Solution. Define

$$h(z) = -if(z).$$

If $f(x)$ is pure imaginary on the segment of the real axis, then $h(x)$ is real there. Since h is analytic in the same domain, the reflection principle applied to h gives

$$\overline{h(\overline{z})} = h(z).$$

Substituting $h = -if$,

$$i\,\overline{f(\overline{z})} = -if(z),$$

so

$$\boxed{\overline{f(\overline{z})} = -f(z)}.$$

Conversely, if this identity holds, then for real x,

$$\overline{f(x)} = -f(x),$$

which means that the real part of $f(x)$ is zero. Hence $f(x)$ is pure imaginary, as required.

□

Exercise 30.5. Write $|\exp(2z+i)|$ and $|\exp(iz^2)|$ in terms of x and y. Then show that

$$|\exp(2z+i) + \exp(iz^2)| \le e^{2x} + e^{-2xy}.$$

Solution. For any complex number w,

$$|e^w| = e^{\operatorname{Re} w}.$$

Since $z = x + iy$,

$$\operatorname{Re}(2z+i) = 2x,$$

so

$$|\exp(2z+i)| = e^{2x}.$$

Also,

$$z^2 = x^2 - y^2 + 2xyi,$$

and hence

$$iz^2 = -2xy + i(x^2 - y^2).$$

Therefore

$$|\exp(iz^2)| = e^{-2xy}.$$

The triangle inequality now gives

$$\boxed{|\exp(2z+i) + \exp(iz^2)| \le e^{2x} + e^{-2xy}}.$$

□

Exercise 30.13. Let the function $f(z) = u(x, y) + iv(x, y)$ be analytic in some domain D. State why the functions

$$U(x, y) = e^{u(x,y)} \cos v(x, y), \qquad V(x, y) = e^{u(x,y)} \sin v(x, y)$$

are harmonic in D.

Solution. Since f is analytic in D and the exponential function is entire, the composition $e^{f(z)}$ is analytic in D. Moreover,

$$\begin{aligned} e^{f(z)} &= e^{u+iv} \\ &= e^u(\cos v + i \sin v) \\ &= U + iV. \end{aligned}$$

By the theorem in Sec. 27, the real and imaginary components of an analytic function are harmonic. Therefore U and V are harmonic throughout D.

□

Exercise 33.2. Show that

(a) $\log e = 1 + 2n\pi i$, $n = 0, \pm 1, \pm 2, \ldots$;

(b) $\log i = \left(2n + \dfrac{1}{2}\right)\pi i$, $n = 0, \pm 1, \pm 2, \ldots$;

(c) $\log(-1 + \sqrt{3}\, i) = \ln 2 + 2\left(n + \dfrac{1}{3}\right)\pi i$, $n = 0, \pm 1, \pm 2, \ldots$.

Solution. For $z \neq 0$, the multivalued logarithm is the set

$$\log z = \{\ln |z| + i(\operatorname{Arg} z + 2n\pi) \mid n \in \mathbb{Z}\}.$$

(a) Since $|e| = e$ and $\operatorname{Arg} e = 0$,

$$\boxed{\log e = \{1 + 2n\pi i \mid n \in \mathbb{Z}\}}.$$

(b) Since $|i| = 1$ and $\operatorname{Arg} i = \pi/2$,

$$\boxed{\log i = \left\{\left(2n + \frac{1}{2}\right)\pi i \mid n \in \mathbb{Z}\right\}}.$$

(c) The modulus of $-1 + \sqrt{3}\, i$ is 2, and its principal argument is $2\pi/3$. Hence

$$\begin{aligned} \log(-1 + \sqrt{3}\, i) &= \left\{\ln 2 + i\left(\frac{2\pi}{3} + 2n\pi\right) \mid n \in \mathbb{Z}\right\} \\ &= \boxed{\left\{\ln 2 + 2\left(n + \frac{1}{3}\right)\pi i \mid n \in \mathbb{Z}\right\}}. \end{aligned}$$

Each set lists precisely the values parameterized by $n = 0, \pm 1, \pm 2, \ldots$ in the exercise.

□

Exercise 33.4. Show that

$$\log(i^2) \neq 2 \log i$$

when the branch

$$\log z = \ln r + i\theta \qquad \left(r > 0, \ \frac{3\pi}{4} < \theta < \frac{11\pi}{4}\right)$$

is used. (Compare this with the example in Sec. 33.)

Solution. Let g denote the single-valued logarithmic branch specified in the exercise, written as log in the textbook. For i, the arguments are $\pi/2 + 2k\pi$. The unique one in the branch interval

$$\frac{3\pi}{4} < \theta < \frac{11\pi}{4}$$

is $5\pi/2$. Hence

$$g(i) = \frac{5\pi}{2}i, \qquad 2g(i) = 5\pi i.$$

On the other hand, $i^2 = -1$, and the unique argument of -1 in the same interval is π. Thus

$$g(i^2) = g(-1) = \pi i.$$

Therefore

$$\boxed{g(i^2) = \pi i \neq 5\pi i = 2g(i)}.$$

□

Exercise 33.11. Show in two ways that the function

$$\ln(x^2 + y^2)$$

is harmonic in every domain that does not contain the origin.

Solution. Let

$$H(x, y) = \ln(x^2 + y^2).$$

First, direct differentiation gives

$$H_x = \frac{2x}{x^2 + y^2}, \qquad H_y = \frac{2y}{x^2 + y^2},$$

and, for $(x, y) \neq (0, 0)$,

$$H_{xx} = \frac{2(y^2 - x^2)}{(x^2 + y^2)^2}, \qquad H_{yy} = \frac{2(x^2 - y^2)}{(x^2 + y^2)^2}.$$

Hence

$$H_{xx} + H_{yy} = 0.$$

All first- and second-order partial derivatives are continuous away from the origin, so H is harmonic there.

For a second argument, fix any $z_0 \neq 0$. A sufficiently small disk about z_0 avoids the origin and admits an analytic branch g of the logarithm. For a continuous choice $\theta(z)$ of argument on that disk,

$$g(z) = \ln|z| + i\theta(z), \qquad \operatorname{Re} g(z) = \ln|z| = \frac{1}{2}\ln(x^2 + y^2).$$

By the theorem in Sec. 27, the real part of an analytic function is harmonic. Consequently $H = 2\operatorname{Re} g$ is harmonic near z_0. Since z_0 was arbitrary, H is harmonic throughout every domain that excludes the origin; a global logarithm branch on that domain is not required.

□

Exercise 34.1. Show that for any two nonzero complex numbers z_1 and z_2,

$$\operatorname{Log}(z_1 z_2) = \operatorname{Log} z_1 + \operatorname{Log} z_2 + 2N\pi i,$$

where N has one of the values $0, \pm 1$. (Compare with Example 2 in Sec. 34.)

Solution. Let

$$\theta_1 = \operatorname{Arg} z_1, \qquad \theta_2 = \operatorname{Arg} z_2, \qquad S = \theta_1 + \theta_2.$$

Since $-\pi < \theta_j \le \pi$,

$$-2\pi < S \le 2\pi.$$

The principal argument of the product is obtained by bringing S into $(-\pi, \pi]$:

$$\operatorname{Arg}(z_1 z_2) = S + 2N\pi,$$

where

$$N = \begin{cases} -1, & S > \pi, \\ 0, & -\pi < S \le \pi, \\ 1, & S \le -\pi. \end{cases}$$

Thus $N \in \{0, \pm 1\}$. Since $|z_1 z_2| = |z_1||z_2|$,

$$\begin{aligned} \operatorname{Log}(z_1 z_2) &= \ln|z_1 z_2| + i \operatorname{Arg}(z_1 z_2) \\ &= \ln|z_1| + \ln|z_2| + i(\theta_1 + \theta_2 + 2N\pi) \\ &= \boxed{\operatorname{Log} z_1 + \operatorname{Log} z_2 + 2N\pi i}. \end{aligned}$$

□

Exercise 34.2. Verify expression (4), Sec. 34, for $\log(z_1/z_2)$ by

(a) using the fact that $\arg(z_1/z_2) = \arg z_1 - \arg z_2$ (Sec. 9);

(b) showing that $\log(1/z) = -\log z$, $z \neq 0$, in the sense that $\log(1/z)$ and $-\log z$ have the same set of values, and then referring to expression (1), Sec. 34, for $\log(z_1 z_2)$.

Solution. The identities here are identities of multiple-valued logarithms, so equality means equality of the corresponding sets of values, with compatible values selected.

(a) Choose arguments $\theta_1 \in \arg z_1$ and $\theta_2 \in \arg z_2$. The argument identity in Sec. 9 states that all arguments of z_1/z_2 have the form $\theta_1 - \theta_2 + 2k\pi$, where $k \in \mathbb{Z}$. Therefore all logarithmic values of the quotient are

$$\ln \frac{|z_1|}{|z_2|} + i(\theta_1 - \theta_2 + 2k\pi), \qquad k \in \mathbb{Z}.$$

These are precisely the differences of the values $\ln|z_1| + i(\theta_1 + 2m\pi)$ and $\ln|z_2| + i(\theta_2 + 2n\pi)$, because $m - n$ ranges over $\mathbb{Z}$. Hence, as sets of values,

$$\boxed{\log\left(\frac{z_1}{z_2}\right) = \log z_1 - \log z_2}.$$

(b) Write $z = re^{i\theta}$. The values of $\log(1/z)$ are

$$-\ln r + i(-\theta + 2k\pi), \qquad k \in \mathbb{Z}.$$

The values of $-\log z$ are

$$-\ln r - i(\theta + 2n\pi) = -\ln r + i(-\theta - 2n\pi), \qquad n \in \mathbb{Z}.$$

These are the same set of values, so

$$\log(1/z) = -\log z.$$

Since

$$\frac{z_1}{z_2} = z_1 \frac{1}{z_2},$$

expression (1), Sec. 34, gives

$$\log\left(\frac{z_1}{z_2}\right) = \log z_1 + \log(1/z_2) = \boxed{\log z_1 - \log z_2}.$$

□

Exercise 34.5. Let z denote any nonzero complex number, written

$$z = re^{i\Theta} \qquad (-\pi < \Theta \leq \pi),$$

and let n denote any fixed positive integer. Show that all of the values of $\log(z^{1/n})$ are given by

$$\log(z^{1/n}) = \frac{1}{n}\ln r + i\frac{\Theta + 2(pn+k)\pi}{n},$$

where $p = 0, \pm 1, \pm 2, \ldots$ and $k = 0, 1, 2, \ldots, n-1$. Then, after writing

$$\frac{1}{n}\log z = \frac{1}{n}\ln r + i\frac{\Theta + 2q\pi}{n}, \qquad q = 0, \pm 1, \pm 2, \ldots,$$

show that the set of values of $\log(z^{1/n})$ is the same as the set of values of $(1/n)\log z$. Thus show that

$$\log(z^{1/n}) = \frac{1}{n}\log z,$$

where, corresponding to a value of $\log(z^{1/n})$ taken on the left, the appropriate value of $\log z$ is to be selected on the right, and conversely. (The result in Exercise 5, Sec. 33, is a special case of this one.)

Suggestion. Use the fact that the remainder upon dividing an integer by a positive integer n is always an integer between 0 and $n-1$, inclusive; that is, any integer q can be written $q = pn + k$, where $p \in \mathbb{Z}$ and $k \in \{0, 1, \ldots, n-1\}$.

Solution. The n distinct values of $z^{1/n}$ are

$$w_k = r^{1/n}\exp\left(i\frac{\Theta + 2k\pi}{n}\right), \qquad k = 0, 1, \ldots, n-1.$$

For each k, the logarithmic values of w_k form the set

$$\begin{aligned}\log w_k &= \left\{\frac{1}{n}\ln r + i\left(\frac{\Theta + 2k\pi}{n} + 2p\pi\right) \mid p \in \mathbb{Z}\right\} \\ &= \left\{\frac{1}{n}\ln r + i\frac{\Theta + 2(pn+k)\pi}{n} \mid p \in \mathbb{Z}\right\}.\end{aligned}$$

Taking every nth root and every logarithmic value, and comparing with the values of $(1/n)\log z$, gives

$$\log(z^{1/n}) = \bigcup_{k=0}^{n-1} \log w_k = \left\{\frac{1}{n}\ln r + i\frac{\Theta + 2(pn+k)\pi}{n} \;\middle|\; p \in \mathbb{Z},\ k \in \{0, 1, \ldots, n-1\}\right\},$$

$$\frac{1}{n}\log z = \left\{\frac{1}{n}\ln r + i\frac{\Theta + 2q\pi}{n} \mid q \in \mathbb{Z}\right\}.$$

By the division algorithm, every integer q has a unique representation $q = pn + k$ with $p \in \mathbb{Z}$ and $0 \leq k \leq n-1$. Thus these sets contain exactly the same values, proving

$$\boxed{\log(z^{1/n}) = \frac{1}{n}\log z}.$$

□